

XINYI ZHAO

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TECHNICAL SKILLS

Hardware: PCB Design (Altium, KiCad), Power Electronics (Buck/Boost, LDO, PMIC), Simulation (LTspice), Wire Harnessing, PCB Manufacturing & Rework, THT/SMD Soldering

Hardware Tools: Oscilloscope, E-Load, Power Supply, Function Generator, DMM, Hi-Pot Testing, Boundary Scan (JTAG), LabVIEW, NI TestStand, Verilog, FPGA (ModelSim, Quartus)

Software: C/C++, Python, Assembly, MATLAB/SimuLink, Git, Jira/Confluence

Communication Protocols: CAN, UART, I2C, SPI, isoSPI, USB, SWD

EDUCATION

4th Year Electrical Engineering – University of British Columbia

2023 – 2028

Bachelor of Applied Science

EXPERIENCE

Electrical Engineering Intern

May 2026 – Present

Kardium

Burnaby, BC

- Owned the end-to-end proposal, design, and validation of production-level test processes for medical-device PCBAs for atrial fibrillation treatment, integrating boundary-scan (JTAG), lab equipment, LabVIEW, and TestStand to achieve over 86% test coverage, enabling earlier detection of assembly defects.
- Designed a custom, multipurpose test platform PCBA to improve boundary-scan and probe access, added current sensing reliability, streamlining production testing of a new board while also supporting standalone functional testing, debugging, and expanded measurement access.
- Developed boundary-scan tests to detect solder-joint faults, alongside functional tests to verify IC operation, including sequenced testing of LEDs, I2C, SPI, flash and memory devices, and FPGA programming.
- Integrated TestStand and LabVIEW with boundary-scan software to create unified, automated test procedures and determining production test procedures.
- Evaluated boundary-scan software, including ProVision, XJDeveloper, and ScanExpress for production deployment by prototyping test sequences, as well as managing vendor communication.
- Supported supply chain and component lifecycle management by reviewing part availability and obsolescence risks, identifying and testing alternative components.

Electrical Engineering Intern

May 2025 – Aug. 2025

ENVO Drive Systems

Burnaby, BC

- Drove electrical system design within the R&D team for the UPT project, a fully electric low-speed vehicle.
- Owned the design of the vehicle's electrical harness to integrate sensors, hub motors, power distribution, and ECUs with mechanical systems and battery. Designed wiring and block diagrams of the high-level electrical system architecture.
- Designed schematics using KiCad, and developed test circuits with ESP32 microcontrollers to collect and verify hub motor thermistor data, validating sensor accuracy and reliability.
- Performed FMEA on the operation of the vehicle harness, guiding component selection and mechanical integration.

PROJECTS

Electrical Technical Director | *UBC Formula Electric FSAE Team*

Sept. 2024 – Present

- Leading a team of 15 EE students to design 6 custom ECUs and 11 supporting PCBs for our electric FSAE racecar's fully custom CAN-networked low voltage electrical system, including the LV power distribution, data and sensor acquisition, DC/DC converter, and BMS.
- Owned the design and bring-up of a smart debug spud in Altium Designer, with an integrated 5V-24V boost converter, e-fuses, and discrete logic, implementing redundant hardware fault protection to address every single point of failure, including shorts and operator errors.
- Validated and debugged the vehicle controller, an 850W low-voltage power-distribution PCB that combines ideal-diode-ORED power sources, and distributes protected power through e-fused outputs to the vehicle's LV subsystems.
- Lead system-level integration and validation of the vehicle's low- and high-voltage electrical systems, resolving cross-functional hardware, firmware, and mechanical interface issues and leading debugging of critical BMS electronics.
- Designed a PCB to validate BMS algorithms using 18650 cells to emulate a battery module, enabling firmware and hardware-level testing of custom BMS daughter PCBs in isolation.
- Mentoring junior electrical engineering students in electronics design and engineering processes by reviewing schematics and PCB layouts, providing technical guidance, and supporting project planning and timeline management.
- Lead design and manufacturing of the vehicle's electrical harness, material selection, and modeling with RapidHarness.